

# Data Quality Log

## Olist E-commerce Delivery & Customer Satisfaction Analysis

This document tracks every data-quality decision made across the Excel (Power Query), SQL (SQLite), and Power BI stages of this project, so the analysis is fully reproducible and auditable.

| Stage     | Issue                                                                    | Count Affected            | Decision & Reasoning                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excel     | order_approved_at missing                                                | 160                       | Left blank — does not affect delay calculation                                                                                                                          |
| Excel     | order_delivered_carrier_date missing                                     | 1,783                     | Left blank — mostly canceled/unavailable/processing orders                                                                                                              |
| Excel     | order_delivered_customer_date missing                                    | 2,965                     | Left blank — delay marked as N/A, order never delivered                                                                                                                 |
| Excel     | customer_date earlier than carrier_date (logical anomaly)                | 23                        | Flagged via Data_Anomaly_Flag column, not deleted — likely a data-entry error at source                                                                                 |
| Excel/SQL | Duplicate reviews per order                                              | 559 rows (555 orders)     | Kept only the most recent review per order (by review_answer_timestamp)                                                                                                 |
| SQL       | product_category_name missing                                            | 610 products              | Used COALESCE(...,'unknown') in revenue-by-category query                                                                                                               |
| SQL       | "Late" definition: full timestamp vs. calendar date                      | 1,292 orders reclassified | Adopted DATE-based definition — same-day delivery counts as on-time, since the estimated date carries no time component                                                 |
| SQL       | Months with very small order counts (e.g. Sept 2016 = 1 order)           | N/A                       | Excluded from "worst month" interpretation (min. 30 orders threshold) to avoid statistically misleading rates                                                           |
| Power BI  | Seller avg. delay: item-level (SQL) vs. order-level (Power BI) averaging | Minor value differences   | Documented as expected — SQL averages per order-item row, Power BI averages per distinct order                                                                          |
| Power BI  | Category filter not cross-filtering KPIs                                 | —                         | Rebuilt data model as a star schema (raw order_items/products/sellers tables) with a bidirectional relationship, instead of relying on pre-aggregated SQL result tables |

All SQL queries are documented individually in /sql/ with inline comments explaining each business question and technical decision.